

The sticky Kakeya conjecture in dimensions five and higher

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November 16, 2025

Abstract. A Kakeya set is defined as a compact subset of $\mathbb{R}^n$ which contains a line segment of length 1 in every direction. The Kakeya conjecture says that every Kakeya set has Minkowski and Hausdorff dimensions equal to n . There is a special class of Kakeya sets called sticky Kakeya sets. Sticky Kakeya sets exhibit an approximate self-similarity at many scales, and sets of this type played an important role in Wang and Zahl's recent resolution of the Kakeya conjecture in $\mathbb{R}^3$. In this talk, I will discuss the motivation for thinking about sticky sets and give an improved Hausdorff dimension estimate for sticky Kakeya sets in higher dimensions.